

# Anesa Ibrahimi

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## SUMMARY

Computer Vision & Machine Learning Engineer with an MSc in Artificial Intelligence. Hands-on industry experience developing digital pathology pipelines for whole-slide image processing, tissue segmentation, stain classification, dataset construction, model validation, and automated quality control. Experienced with Python, PyTorch, OpenCV, large-scale image processing, and GPU/HPC experimentation, with published research in vision-language models and multimodal AI.

## TECHNICAL SKILLS

**Languages:** Python, SQL, MATLAB  
**Machine Learning / Deep Learning:** PyTorch, TensorFlow, HuggingFace Transformers, Keras  
**Tools & Infrastructure:** Docker, Git, CUDA, Slurm (HPC), Linux, Jupyter, 3D Slicer  
**Specializations:** Computer Vision, Medical Imaging, Multimodal AI, Vision-Language Models, Information Retrieval, Reinforcement Learning

## EDUCATION

- Universiteit van Amsterdam** 2023-2025  
*MSc. Artificial Intelligence* Amsterdam, Netherlands
- Vrije Universiteit Amsterdam** 2020-2023  
*BSc. Artificial Intelligence, Specialisation : Intelligent Systems, Minor: Data Science* Amsterdam, Netherlands

## PROFESSIONAL EXPERIENCE

- ARTIDIS AG** May 2026 – Present  
*Computer Vision & MLOps Engineer Intern* Basel, Switzerland
  - Developing **computer vision pipelines for digital pathology and medical imaging**, processing high-resolution whole-slide images across heterogeneous staining protocols and slide formats.
  - Built and refined **tissue segmentation and post-processing pipelines** using connected-component analysis, contour filtering, and noise removal to improve robustness of pathology tissue masks.
  - Developed automated workflows for **CZI/SVS processing, microscopy metadata extraction, thumbnail generation, and visual QC**, enabling systematic validation of large pathology slide collections.
  - Built **dataset and model-validation pipelines** for histopathology classification and segmentation, including annotation harmonization, patch generation, dataset manifests, standardized preprocessing, stain classification, and unseen-slide evaluation.
  - Technologies:** Python, PyTorch, OpenCV, NumPy, Pandas, OpenSlide, Git, Digital Pathology, Whole-Slide Imaging.

## ACADEMIC EXPERIENCE

- Vrije Universiteit Amsterdam** March 2023 - May 2023  
*Professor's Assistant - Databases* Amsterdam, Netherlands  
*Professor: Dr. J. Endrullis*
  - Delivered weekly database labs to ~ 50 undergraduate students, covering relational modeling and SQL query optimization.
  - Mentored students on relational schema design and SQL queries, supporting project completion across multiple assignments.
- Vrije Universiteit Amsterdam** March 2022 - May 2022  
*Professor's Assistant – Human-Computer Interaction* Amsterdam, Netherlands  
*Professor: Dr. E.M.A.L. Beauxis Aussalet*
  - Led weekly HCI practical sessions for students covering interactive system design and prototyping workflows.
  - Guided teams in applying usability testing and evaluation methods for interactive systems.
  - Provided technical feedback on project prototypes and UX evaluation reports.

## TECHNICAL PROJECTS

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### • Fine-grained image understanding with Vision-Language Models

January 2025 - August 2025

*Master Thesis Project*

- Designed and implemented a novel **Memory-Augmented Vision Encoder** by integrating trainable key-value memory modules into the MLP layers of a Vision Transformer (ViT).
- Developed a two-stage training pipeline leveraging **knowledge distillation** to transfer representations from a pre-trained CLIP (teacher) to the memory-enhanced student model, enabling efficient training.
- Achieved state-of-the-art performance, decisively outperforming the baseline CLIP model on **long-caption, fine-grained retrieval benchmarks**.
- Trained and evaluated on HPC infrastructure across **Snellius** (4× NVIDIA A100/H100) and **DAS-5** partitions using Slurm for multi-GPU data parallel experiments.
- **Technologies:** Python, PyTorch, CUDA, OpenAI CLIP, Vision Transformers, Slurm/HPC (Snellius, DAS-5).

### • Multi-Modal NAO Robot

October 2024 - December 2024

- Led a 7-member team in developing a **multi-modal emotion recognition and response system** for the NAO robot.
- Engineered a distributed multimodal perception pipeline combining RoBERTa (text), SpeechBrain (audio), and LLaMA Vision models with OpenVoice emotional speech synthesis.
- Conducted controlled human-robot interaction experiments with **10 participants and 20 interaction trials**, achieving **60%** exact emotion classification accuracy and **90%** accuracy at emotional category level.
- **Technologies:** Python, PyTorch, Docker, OpenVoice, HuggingFace models, NAO Framework.

### • LLM4CS

October 2024 - December 2024

- Reproduced and extended the LLM4CS framework to evaluate large language models as **context-aware search intent interpreters** in multi-turn conversational search tasks.
- Benchmarked prompting strategies and response aggregation methods across **dense (Faiss) and sparse (BM25) retrieval pipelines** using Pyserini.
- Analyzed retrieval effectiveness of LLM-assisted conversational search pipelines across multiple query reformulation strategies.
- **Technologies:** Python, Pyserini, Information Retrieval, LLMs, Prompt Engineering, Dense & Sparse Retrieval.

### • AI for Medical Imaging (AI4MI) – SegTHOR Challenge

September 2024 - October 2024

- Developed CNN-based models for segmenting **thoracic organs-at-risk** (esophagus, heart, trachea, aorta) from 3D CT scans in the SegTHOR dataset (60 patients).
- Improved ENet segmentation performance through loss-function optimization, data augmentation, and regularization tuning.
- Benchmarked multiple architectures (ENet, UNet, UNet++, DeepLabV3+) using **Dice, Jaccard, Hausdorff Distance, Precision/Recall**.
- **Technologies:** Python, PyTorch, Medical Image Segmentation, 3D Slicer, CNN Architectures.

### • In-Context Learning Improves Compositional Understanding of Vision-Language Models

April 2024 - May 2024

*Published at ICML 2024 Workshop on Foundation Models in the Wild*

- Conducted a systematic benchmark of contrastive (CLIP-based) and generative VLMs across compositional reasoning datasets including Winoground, ARO, and SugarCreme.
- Designed an in-context learning (ICL) prompting pipeline using synthetic and COCO images, improving compositional reasoning performance by up to **~6% on ARO benchmarks**.
- First-authored a **peer-reviewed workshop paper at ICML 2024**, presenting insights into architectural limitations and prompting strategies for vision-language compositionality.
- **Technologies:** Python, PyTorch, Vision-Language Models, CLIP, LLaVA, CogVLM, In-Context Learning

### • Instance Diffusion - Extension

April 2024 - May 2024

- Reproduced the InstanceDiffusion framework for instance-controlled image generation using the MSCOCO dataset and diffusion-based models.
- Developed an LLM-powered module to automatically generate image captions and bounding boxes, eliminating manual annotation in the generation pipeline.
- Improved bounding-box detection performance in reproduction experiments.
- **Technologies:** Python, PyTorch, Diffusion Models, Stable Diffusion, CLIP, MSCOCO

### • Optimizing Locomotion with Evolutionary Algorithms in MuJoCo

March 2023 - June 2023

*Bachelor Thesis Project – Vrije Universiteit Amsterdam*

- Explored the potential of Evolutionary Algorithms (CMA-ES, XNES, SNES) for solving continuous control optimization problems in the MuJoCo Ant environment.
- Designed and implemented a full simulation and optimization pipeline for evolving the ant agent's locomotion behavior across 100 virtual environments with varying terrain densities and obstacle heights.
- Conducted quantitative and qualitative analyses to benchmark the performance of each algorithm.
- **Technologies:** Python, MuJoCo, OpenAI Gym, EvoTorch, NumPy, Pandas, Matplotlib.

## PUBLICATIONS

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•**Anesa Ibrahimi**, Matteo Nulli, Avik Pal, Hoshe Lee, Ivona Najdenkoska, *In-Context Learning Improves Compositional Understanding of Vision-Language Models*, arXiv, 2024. <https://arxiv.org/abs/2407.15487>

–Paper published at ICML Foundation Models in the Wild Workshop